Imaging around corners via transient Gaussian field

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***Abstract*—The ability to imaging hidden objects or non-line-of-sight (NLOS) imaging represents a frontier in imaging science and computer vision. Despite remarkable advances, these approaches rely on the rigid discretizations of voxel representations, which necessitate dense scanning, long collection times, and limited aperture. Consequently, NLOS imaging remain distant from real-world applications. Here we report the paradigm of transient Gaussian field to resolve the long-standing limitations of voxel reconstruction. By coupling continuous Gaussian field with transient volume rendering, our framework compresses the parameter space by orders of magnitude, enabling high-fidelity 3D reconstruction in real time. Furthermore, the Gaussian field expands the accessible spatial-frequency support, thereby overcoming the conventional finite-aperture constraint. In experiment, we demonstrate the first 3D panoramic NLOS imaging with unprecedented scene completeness and angular coverage. Our work establishes a new benchmark for NLOS imaging and moves it decisively closer to practical deployment.**

***Index Terms*—non-line-of-sight imaging, transient Gaussian field, 3D panorama, noise robustness, sparse reconstruction**

I. Introduction

NON-line-of-sight (NLOS) imaging [1], [2] aims to recover scenes hidden from direct view by analyzing multiply scattered light, offering the ability to "see around corners" through indirect optical measurements. While various experimental techniques including speckle correlations [3]–[5], wavefront shaping [6], and coherent detection [7], [8] have been explored for NLOS imaging, time-of-flight (ToF) based approaches [9]–[19] have established themselves as the most robust solution for real-world scenarios. By illuminating a relay surface and recording the arrival times of returning photons, ToF systems acquire transient measurements that encode the hidden three-dimensional (3D) geometry. Recovering the hidden scene from these sparse and noisy transients, however, remains a severely ill-posed inverse problem.

The solution to this problem is critically dependent on how the hidden scene is represented. Most existing NLOS reconstruction pipelines are built upon explicit voxel grids [9]–[15], [20]. While this representation has driven much of the early progress in NLOS imaging [9]–[12] and enabled elegant linear formulations [10], it faces a severe performance plateau. First, the parameter space of voxel grids scales cubically ($O(N^3)$) with the grid size N. Such high dimensionality necessitates dense spatial scanning [11] and prolonged photon collection [12] to ensure stable and accurate inversion. Moreover, voxel-based approaches are constrained by the finite-sized sampling aperture. They can only reconstruct surfaces facing the detection area, making reconstructed geometry fragmented and discontinuous [21]. Recent efforts to transcend voxel representations using implicit neural fields [22]–[24], triangular meshes [25], and radial basis functions [26] improve representational flexibility, yet often incur heavy computational overhead, slow optimization convergence, and poor noise robustness, remaining difficult to generalize to real-world NLOS scenarios.

Here we introduce transient Gaussian field, a continuous and spatially adaptive representation for transient NLOS imaging based on 3D Gaussian primitives [27]–[29]. Rather than discretizing hidden geometry into independent volumetric cells, our framework models the hidden scene as a continuous Gaussian field jointly optimized through transient volume rendering. This formulation fundamentally alters the recoverable information structure of NLOS imaging. We show that Gaussian field provides a highly compact and noise-robust parameterization of hidden scenes. The effective representation complexity empirically scales from the $O(N^3)$ behavior of voxel grids toward an $O(N^2)$ regime, reducing the measurement burden required for stable reconstruction. Moreover, the intrinsic smoothness and spatial adaptivity of Gaussian primitives naturally regularize the inverse problem and improve robustness. Furthermore, continuous transient rendering expands the effective spatial-frequency support recoverable from finite-aperture measurements, overcoming the limitation that constrains existing approaches to fragmented and orientation-dependent reconstructions. Leveraging these capabilities, we present the first experimental demonstrations of 3D panoramic NLOS imaging, recovering the full 360° geometry of hidden scenes with unprecedented structural completeness. We showcase reconstruction results on complex, unengineered everyday scenarios that were previously entirely intractable.

To date, NLOS imaging has been largely confined to recovering the intensity and depth of surfaces visible to a finite relay aperture [3]–[21], [30]–[34]. By expanding the effective spatial-frequency support recoverable from transient measurements, our transient Gaussian field enables 3D panoramic

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recovery of scene geometry for the first time. This capability further enables free-viewpoint navigation within the reconstructed hidden volume and opens new opportunities for applications requiring comprehensive spatial awareness, including disaster response and intelligent robotic perception.

II. Related Work

A. NLOS Imaging Algorithms

Existing ToF-based NLOS reconstruction methods can be broadly categorized by their underlying 3D scene representations, primarily divided into explicit volumetric grids and implicit neural representations.

Voxel-based techniques [9]–[15] discretize the hidden scene into rigid 3D volumetric grids and reconstruct the geometry by solving a transient inverse problem. Among these, the light-cone transform (LCT) [10] establishes a computationally efficient framework by reformulating confocal transient measurements as a 3D convolution, thereby enabling rapid frequency-domain filtering. Phasor field [21] interprets NLOS measurements through the lens of virtual wave propagation, allowing for the imaging of complex hidden scenes. While mathematically elegant, these explicit formulations suffer from a cubically scaling parameter space, which necessitates dense spatial scanning and prolonged photon acquisition to ensure stable inversion. Furthermore, due to the finite aperture of the relay wall, the reconstruction is inherently constrained to surfaces oriented toward the visible boundary, inevitably resulting in discontinuous geometry [21]. Beyond these analytical approaches, the voxelized representation naturally pairs with data-driven frameworks. Learning-based architectures, such as NLOST [35] and LFE [36], typically leverage learned priors to directly regress 3D volumes. Although these methods excel in sparse reconstruction [37] and exhibit localized noise robustness [35] in specific contexts, they struggle to generalize across varying experimental setups and novel scenes, thereby undermining their practical utility.

Implicit neural representations have recently emerged to circumvent the discretization constraints inherent in voxel grids. For instance, the neural transient field (NeTF) [22] employs multi-layer perceptrons (MLPs) to model the hidden volume continuously, rendering transient histograms via differentiable volume integration. However, NeTF relies on stochastic ray sampling across angular dimensions, which introduces heavy computational overhead, slow convergence, and extreme sensitivity to experimental noise. Other geometric primitives, such as triangular meshes [25] or radial basis functions [26], similarly require solving highly non-convex optimization problems to recover the hidden structure. Consequently, they suffer from prohibitive optimization latency and fragile noise tolerance, posing severe deployment challenges in real-world scenarios.

Our transient Gaussian field departs from these paradigms by pioneering the integration of 3D Gaussians into the domain of NLOS imaging. While categorized as an explicit representation, it uniquely inherits the sparsity and continuity typically reserved for implicit neural representations. Leveraging our tailored rendering and optimization pipeline, the proposed framework is highly versatile, seamlessly accommodating both confocal and non-confocal configurations while exhibiting exceptional reconstruction fidelity under sparse scanning and severe noise. Notably, our approach significantly expands the spatial-frequency support, unlocking 3D panoramic NLOS imaging—a capability that remains unattainable for prior voxelized or implicit formulations.

B. 3D Gaussian Splatting

3D Gaussian Splatting (3DGS) [27] has recently revolutionized 3D computer vision by introducing an explicit point-based scene representation. Unlike neural radiance fields (NeRF) [38] that implicitly encode scenes within neural network weights, 3DGS models a scene as a collection of anisotropic 3D Gaussian primitives, each characterized by position, covariance, opacity, and view-dependent color. Rendering is performed via differentiable tile-based rasterization, achieving real-time performance while maintaining state-of-the-art visual quality. This efficiency and quality have sparked widespread adoption across diverse applications, including dynamic scene reconstruction [39], large-scale environment modeling [40], inverse rendering and relighting [41], physics-based simulation [42], and simultaneous localization and mapping (SLAM) [43]. Comprehensive surveys [28], [29] further document the rapid evolution and expansive breadth of these 3DGS variants.

Despite these successes, existing Gaussian-based methodologies are fundamentally tailored for conventional line-of-sight (LOS) imaging. In these standard regimes, observations comprise direct-view 2D RGB images governed by straight-line ray tracing or standard volumetric rasterization. These formulations rely heavily on the assumptions of explicit camera rays and unoccluded line-of-sight visibility. This makes them inherently incompatible with ToF NLOS measurements, where the hidden geometry is encoded within time-resolved, indirect light transport resulting from multiple diffuse scattering events. Consequently, neither the forward rendering formulation nor the optimization strategies of current 3DGS methods can be straightforwardly adapted to the NLOS inverse problem.

Our work bridges this gap by extending Gaussian representations to transient imaging for the first time. We derive a transient Gaussian forward model directly from the physical light transport process and formulate NLOS reconstruction as an inverse rendering problem over continuous Gaussian fields. Furthermore, to accommodate the severe shot noise and ill-posedness of transient measurements, we develop an octree-guided structural optimization framework that progressively refines the Gaussian field through deterministic subdivision and pruning. These innovations fundamentally distinguish our framework from existing Gaussian-based scene representations developed for conventional computer vision tasks.

III. Principle

A. Forward Model

The principle of optical transient-based NLOS imaging is illustrated in Fig. 1(a). A pulsed laser is directed onto the relay wall, subsequently illuminating the entire hidden volume. Photons scattered by the occluded objects travel back to the wall, where their time-of-flight is precisely recorded

as transient histograms. Let $\boldsymbol{p_l}$ denote the laser illumination point and $\boldsymbol{p_c}$ the detection point on the relay wall. If the detection point is collocated with the laser point ($\boldsymbol{p_l} = \boldsymbol{p_c}$), the configuration is referred to as a confocal setup [10], [17], [18]; otherwise, it is termed a non-confocal setup [9], [11]. Both configurations are standard experimental scenarios employed in optical NLOS imaging.

We represent the hidden scene using a continuous field composed of isotropic 3D Gaussian basis functions $\{G_i\}$. Each individual Gaussian component is described by its position $\boldsymbol{\mu_i}$, scale σ_i, opacity o_i, and reflectance ρ_i. Then, the reflectance field $\rho(\boldsymbol{x})$ at a spatial coordinate $\boldsymbol{x}$ within the hidden volume is modeled as a mixture of these 3D Gaussian distributions,

$$\rho(\boldsymbol{x}) = \sum_i \rho_i f_i(\boldsymbol{x}), f_i(\boldsymbol{x}) = \frac{1}{(2\pi)^{3/2}|\boldsymbol{\Sigma_i}|^{1/2}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu_i})^T \boldsymbol{\Sigma_i}^{-1}(\boldsymbol{x}-\boldsymbol{\mu_i})\right), \quad (1)$$

where $\boldsymbol{\Sigma_i} = \sigma_i^2 \boldsymbol{I}$.

A key challenge in the forward model is mapping this 3D spatial field $\rho(\boldsymbol{x})$ to the ToF measurements. Since the ToF signal corresponds to the photon path length, we must determine the distribution of the Gaussian field mass along the radial distance from the wall. We first derive this relationship under the confocal setting ($\boldsymbol{p_l} = \boldsymbol{p_c}$). By transforming the isotropic 3D Gaussian distribution into spherical coordinates and integrating over the angular components, we obtain its radial profile. This radial distribution is not a simple Gaussian but can be accurately approximated as a weighted sum of a 1D Gaussian and a 1D Rayleigh distribution. We define this unified radial basis function as $K_i(d) = c_i P_{\text{Gaussian}}(d, \sigma_i) + (1-c_i) P_{\text{Rayleigh}}(d, \sigma_i)$, where d represents the distance deviation from the center $\boldsymbol{\mu_i}$ and the coefficient c_i also serves as an optimized parameter. Remarkably, this basis function is applicable to the non-confocal setting as well (see Supplementary Section 1 for details). This allows us to formulate a unified forward model for both scenarios:

$$\tau(\boldsymbol{p_l}, \boldsymbol{p_c}, r) = \sum_i \frac{\rho_i}{(r_i^l r_i^c)^2} \cdot K_i(r - r_i^l - r_i^c) \cdot V(\boldsymbol{\mu_i}; \boldsymbol{p_l}, \boldsymbol{p_c}), \quad (2)$$

where $\tau(\boldsymbol{p_l}, \boldsymbol{p_c}, r)$ is the ToF measurement, and $r_i^l = ||\boldsymbol{p_l} - \boldsymbol{\mu_i}||_2$ and $r_i^c = ||\boldsymbol{p_c} - \boldsymbol{\mu_i}||_2$ are the distance from $\boldsymbol{\mu_i}$ to the laser point $\boldsymbol{p_l}$ and camera detection point $\boldsymbol{p_c}$, respectively. The final factor $V(\boldsymbol{\mu_i}; \boldsymbol{p_l}, \boldsymbol{p_c})$ is the visibility from $\boldsymbol{p_l}$ to $\boldsymbol{\mu_i}$ and then back to $\boldsymbol{p_c}$, which ranges from 0 to 1, and can be calculated by volume rendering.

The volume rendering process is depicted in Fig.1. From each detection point, S rays are emitted into the hidden volume. Due to the continuous and partially transparent nature of the Gaussian field, each ray intersecting a Gaussian component is partially reflected, while the remaining light continues to propagate through the scene. To model this process, intersections along each ray are evaluated sequentially. For the j-th Gaussian element encountered by a ray, its local visibility relative to the illumination point $\boldsymbol{p_l}$ represents the fraction of light that reaches the component without being occluded by previous ones. Then, the global visibility at $\boldsymbol{p_l}$ is obtained by aggregating the local visibility contributions over all S rays:

$$V(\boldsymbol{\mu_i}; \boldsymbol{p_l}, \boldsymbol{p_c}) = \frac{1}{S} \sum_{s=1}^{S} \alpha_i^{(s)} \prod_{j=1}^{k_i^{(s)}-1} \left(1 - \alpha_j^{(s)}\right), \quad (3)$$

where $\alpha_i^{(s)}$ denotes the localized opacity contribution of the i-th Gaussian component along the s-th ray, and $k_i^{(s)}$ represents its intersection order on that ray. The product term $\prod_{j=1}^{k_s-1}(1-\alpha_j^{(s)})$ models the accumulated transmittance before reaching the i-th Gaussian component. Finally, by substituting the computed V_i values into Eq. 2, the 3D Gaussian field can be volumetrically integrated into transient measurements, forming the simulated NLOS response observed at the detection point.

B. Transient Volume Rendering

We now detail the implementation of our volume rendering pipeline. As shown in Supplementary Fig.S2, the spatial extent of the hidden scene is first estimated from the peaks in the time-of-flight measurements and bounded by a spherical enclosure for computational efficiency. Each detection point is modeled as a virtual camera, with associated view and projection matrices. The view matrix transforms coordinates from the detection point's local system to a camera frame oriented toward the center of the spherical bounding volume. The projection matrix is configured as a perspective projection whose field-of-view (FoV) fully encompasses the entire hidden target region from the detection point's viewpoint. Since the target is bounded by a sphere, the FoV can be derived analytically through the trigonometric relation $\text{FoV} = 2\arcsin\left(\frac{r_{\text{sphere}}}{d_{\text{center}}}\right)$, where r_{sphere} is the radius of the bounding sphere and d_{center} denotes the distance from the detection point to the sphere center. With the virtual camera transformations established, each detection point effectively captures the hidden scene through $H \times W$ emitted rays. This formulation allows us to adopt a multi-view volume rendering framework: the localized Gaussian primitives are sorted by depth and blended along each ray according to Eq. 3. The final visibility value is obtained by integrating contributions along the H and W dimensions.

To further reduce both memory usage and runtime, we adopt a tile-based rendering technique. Because each Gaussian component has a limited spatial extent, it typically overlaps with only a small subset of neighboring elements. Thus, we can parallelize the rendering pipeline with smaller tiles (typically 16×16). For each tile, we efficiently identify potentially contributing Gaussian primitives by leveraging their projected 2D bounding boxes and depth ranges. This selection process is optimized with a fast, sorting-based algorithm that minimizes redundant memory access and computation. Each tile is then processed independently by a dedicated thread, where the selected Gaussian primitives are depth-sorted and alpha-blended according to Eq. 3. After all tiles are rendered, their partial results are seamlessly merged to form the final transient image. This divide-and-conquer strategy reduces GPU memory consumption to less than 10 GB and achieves rendering speeds

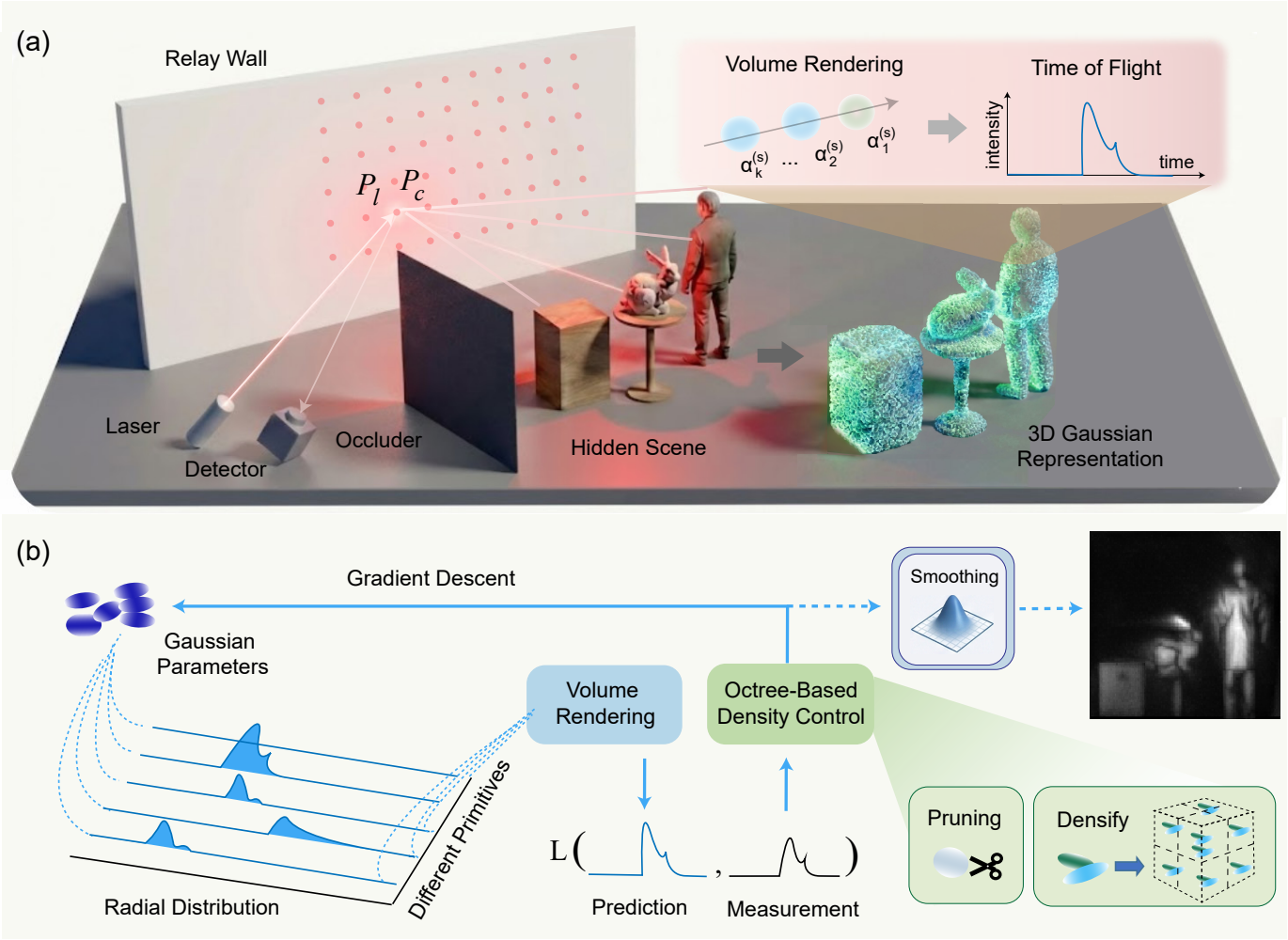


Fig. 1. **The transient Gaussian field framework.** (a) Forward model. The hidden scene is encoded as a continuous Gaussian field, parameterized by a set of isotropic 3D Gaussian primitives. Transient signals are synthesized using a differentiable transient volume rendering model. (b) Reconstruction pipeline. The framework minimizes the discrepancy loss $\mathcal{L}$ between predicted transients and physical single-photon measurements, updating Gaussian parameters iteratively. To faithfully preserve scene geometry and suppress noise, an octree-structured density control mechanism adaptively refines or prunes the Gaussian primitives. The optimized Gaussian field enables seamless free-viewpoint rendering of the hidden scene from arbitrary viewpoints. $\boldsymbol{P_l}$: laser illumination position. $\boldsymbol{P_c}$: detection position. $\alpha_i^{(s)}$: localized opacity contribution of the i-th Gaussian element along the s-th ray.

of several milliseconds, even when dealing with millions of Gaussian primitives. Consequently, our method can scale efficiently to highly complex scenes while maintaining high computational throughput.

IV. Reconstruction

Based on the forward model, we formulate the NLOS reconstruction as an inverse rendering problem, which is solved through maximum likelihood estimation. As illustrated in Fig. 1(b), the hidden volume is represented by an initialized set of 3D Gaussian field components. For each detection point, the corresponding transient histogram is synthesized using the derived forward operator (Eq. 2). To align the prediction with physical measurements, we minimize the discrepancy between the synthesized and captured histograms. Given the photon-starved nature of NLOS imaging, we employ the Poisson negative log-likelihood loss, which is statistically optimal for modeling the shot noise inherent in single-photon detection [23]. The objective function is defined as:

$$L(\hat{\tau}, \tau) = \sum_{r} \left(\hat{\tau}(\boldsymbol{p_l}, \boldsymbol{p_c}, r) - \tau(\boldsymbol{p_l}, \boldsymbol{p_c}, r) \log(\hat{\tau}(\boldsymbol{p_l}, \boldsymbol{p_c}, r))\right), \tag{4}$$

where $\hat{\tau}$ and τ denote the predicted and measured transient histograms, respectively. Throughout the optimization, adaptive densification and pruning are performed at regular intervals to regulate the number of Gaussian primitives. The Gaussian parameters are iteratively updated via gradient descent until convergence.

Though sharing foundational optimization principles with 3D Gaussian representations in computer vision [27], [28], our reconstruction framework adapts these concepts into a unique field framework tailored for NLOS imaging. The prominent shot noise in NLOS transient measurements necessitates a noise-robust optimization pipeline. Octree structures are widely recognized in signal processing for their ability to

enhance signal fidelity and suppress noise by imposing spatial sparsity and regularity [44], [45]. By leveraging this property, our method enforces sparse and structured constraints on the reconstruction field. Instead of allowing individual Gaussians to drift freely, we adopt a coarse-to-fine evolution where the field is deterministically subdivided into octants (eight child Gaussian) within the octree grid. In this process, the positions of the Gaussian primitives are fixed to the node centers and do not serve as optimized parameters. This structural prior effectively acts as a spatial regularizer, preventing the field from fitting to transient noise spikes while preserving the underlying continuous geometry. Besides, we employ a gradient accumulation technique to further stabilize optimization under noisy conditions.

Algorithm 1 Transient Gaussian Field Reconstruction

Input: Transient measurements τ, detection points $\mathcal{P}$
Initialize the Gaussian field $\mathcal{G}$
repeat
  Render transient histograms $\hat{\tau}$ from $\mathcal{G}$
  Compute Poisson loss $\mathcal{L}(\hat{\tau}, \tau)$
  Update Gaussian attributes via gradient descent
  if density control step **then**
    Remove Gaussian nodes with low reflectance
    Expand remaining nodes into eight octree children
  end if
until convergence
Remove residual artifacts
return reconstructed hidden scene

The complete reconstruction pipeline is summarized in Algorithm 1, which consists of four main stages.

(1) Initialization. The transient Gaussian field requires careful initialization of parameters to ensure stable convergence. The Gaussian primitives are initially placed at the centers of the octree root nodes to facilitate subsequent structural density control. The reflectance value ρ_i is initialized to 0.1 and activated through a square activation function. The Gaussian scale σ_i is set relative to the scene dimensions using an exponential activation function to guarantee a positive value. Both the independent optimization coefficient c_i and the opacity parameters are constrained within the range [0, 1] using sigmoid activations and initialized to 0.5. This initialization provides a balanced starting point for optimization and helps stabilize the early training phase.

(2) Loss Calculation and Gradient Descent. The ToF measurements are first normalized by their maximum value to map the signal into the range [0, 1]. During training, a small batch of detection points, typically from 4, 16 to 32, is selected. For each detection point, the current Gaussian field is rendered into transient histograms using the forward model, and a Poisson negative log-likelihood loss is computed between the predicted and measured histograms. To improve optimization stability and reduce memory consumption, we adopt a gradient accumulation strategy: gradients are calculated individually for each detection point and accumulated before performing backpropagation. The Adam optimizer is used, with a learning rate of 0.01 for the coefficient and opacity parameters, and 0.001 for the reflectance ρ and scale parameters.

(3) Density Control Technique. To adaptively refine the spatial profile of the Gaussian field, we introduce a structural density control mechanism that dynamically prunes and adds Gaussian primitives during optimization.

Pruning: Gaussian primitives with reflectance values below a predefined threshold are periodically removed, as these low-reflectance regions usually correspond to empty or non-contributing areas in the hidden volume. This step helps prevent redundant computation and avoids the accumulation of meaningless components.

Adding: To increase spatial resolution in informative regions, each remaining component is deterministically subdivided along the x, y, and z axes into eight child Gaussians, centered within the corresponding subregions. The child Gaussians inherit most parameters from their parent, while their scale σ_i is halved to capture finer local structures. This structural evolution follows the hierarchy of an octree, where the positions of the Gaussian primitives remain strictly fixed to the node centers rather than optimized as free parameters, thereby acting as a powerful geometric regularizer.

(4) Post-processing. After optimization converges, the resulting field representation may still include residual high-frequency noise. We apply two denoising operations to obtain a clean reconstruction. First, components with reflectance values below a threshold (for example, 1×10^{-4}) are removed. Second, Gaussian primitives near the spatial boundaries of the initial reconstruction volume are discarded. The remaining continuous field is then integrated into 2D images following the procedure described in Supplementary Section 1. Since the raw field profiles may appear slightly fragmented, a Gaussian filter is applied to smooth the final 2D images, yielding a coherent and visually stable reconstruction of the hidden scene.

The codes and data are publicly available in github.

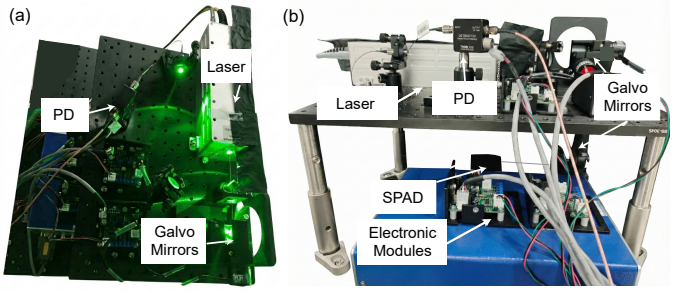


Fig. 2. **Photographs of the experimental NLOS imaging setup.** (a) The top view of the system. (b) The side view of the system. Key components include the laser, galvanometer (galvo) mirrors, and single-photon avalanche diode (SPAD). PD: Photodiode.

V. NLOS Imaging System

To demonstrate NLOS imaging, we have designed a compact and portable platform which supports diverse experimental configurations in this work. As illustrated in Fig.2, the system is integrated on two optical breadboards. This highly compact layout facilitates flexible deployment for probing

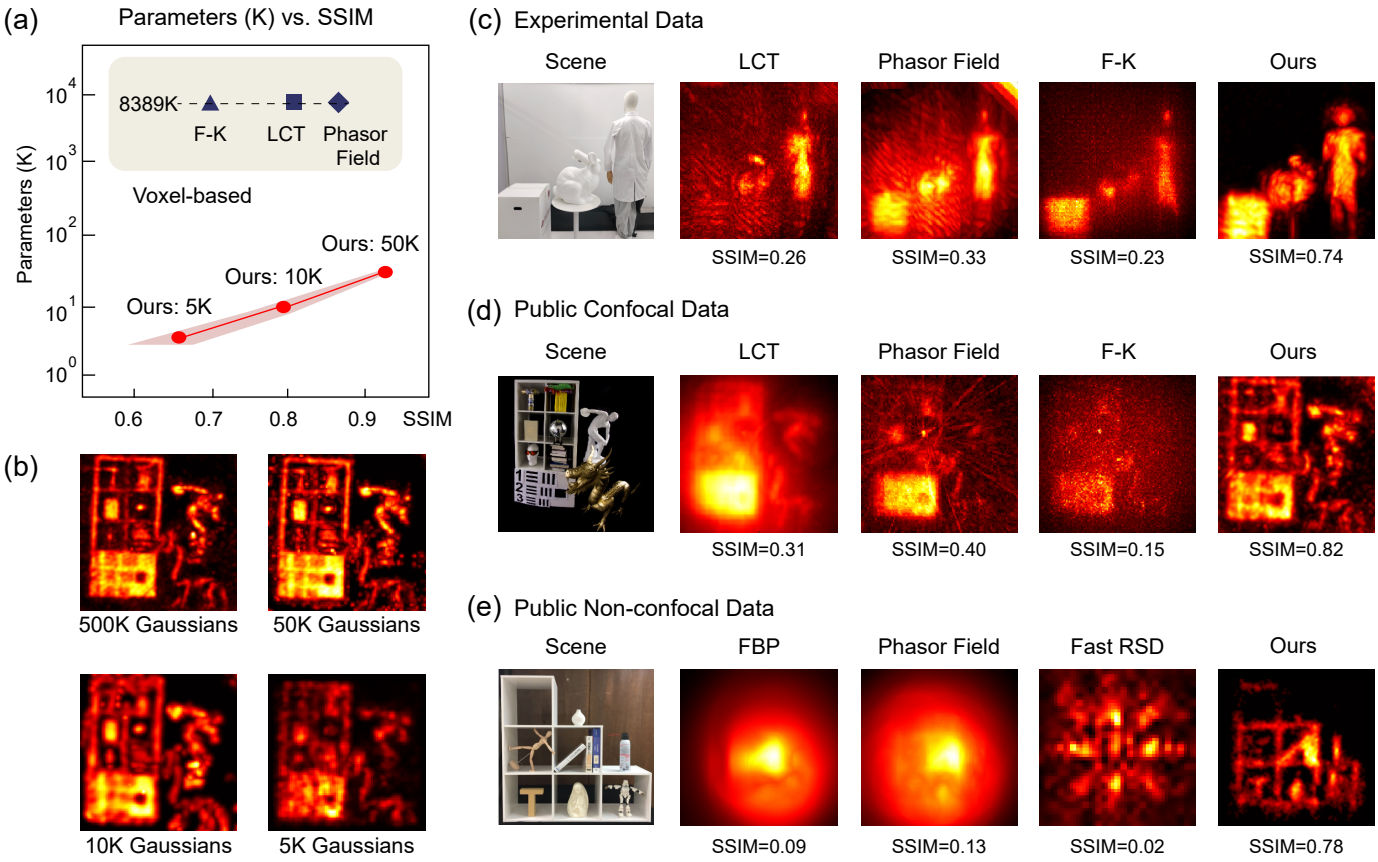


Fig. 3. **Reduced parameter space and robust reconstruction under sparse sampling.** (a-b) The proposed Gaussian field representation significantly compresses the parameter space, achieving competitive SSIM scores with 2-3 orders of magnitude fewer parameters than voxel-based baselines (demonstrated here with 128×128 sampling points). This compact representation enables high-quality reconstruction even under extremely sparse sampling conditions. (c) Reconstructed results on experimental data using 16×16 sampling points. (d-e) Reconstructed results on public confocal [12] and non-confocal [11] datasets, using 24×24 sampling points.

various relay walls, which is particularly critical for our 3D panoramic NLOS experiments. The system employs a pulsed laser source (CNI FL-532 Pico, 400 mW at 532 nm) with an approximate pulse width of 50 ps. Photon detection is performed using a gated single-photon avalanche diode (SPAD, PicoQuant PDM PD-100-CTC-FC). The detector output is recorded by a time-correlated single-photon counting (TCSPC) module (Time Tagger Ultra, Swabian Instruments), which provides a timing resolution of approximately 30 ps and a dead time of 77.4 ns. The laser operates in an external trigger mode synchronized with the TCSPC. The overall temporal resolution of the system is determined at the system level. Contributions from the laser (50 ps), SPAD detector (37 ps), photocell and discriminator electronics (50 ps), TCSPC (8 ps), and a 0.4 m multimode fiber (50 ps) result in a total system temporal resolution around 95 ps.

To maximize acquisition efficiency and signal quality, we adopt a dual-axis scanning strategy. Two galvanometers (Thorlabs GVS012) independently control the laser illumination path and the photon collection path. This configuration allows flexible scanning patterns and reduces the impact of excessively strong first-bounce peaks. Furthermore, we perform rigorous system calibration, specifically targeting the temporal origin and scanning geometry. This step ensures strict consistency between the physical measurements and the forward model used for reconstruction, guaranteeing reliable transient data.

VI. Results

A. Sparse Reconstruction

Voxel-based methods discretize the scene on fixed grids, where the number of parameters scales cubically ($O(N^3)$) with the grid size N. In contrast, our adaptive 3D Gaussian field explicitly parameterizes only occupied areas and conforms tightly to continuous surfaces, causing the number of required components to scale approximately as $O(N^2)$ [46]. Consequently, the solution space becomes orders of magnitude more compact than that of voxel grids. Fig. 3(a)(b) empirically validates this efficiency through a quantitative comparison on the complex "fk-teaser" scene [12] with 128×128 sampling. Our method achieves high-fidelity reconstruction with a structural similarity index measure (SSIM) greater than 0.9 using approximately 5×10^4 components, reducing the parameter space by nearly two orders of magnitude compared to the $128 \times 128 \times 512$ voxels required by standard voxel-based methods. It also maintains the structural integrity of the hidden scene even when the field is compressed to just 5000 elements. Supplementary Fig.S4 provides additional evidence

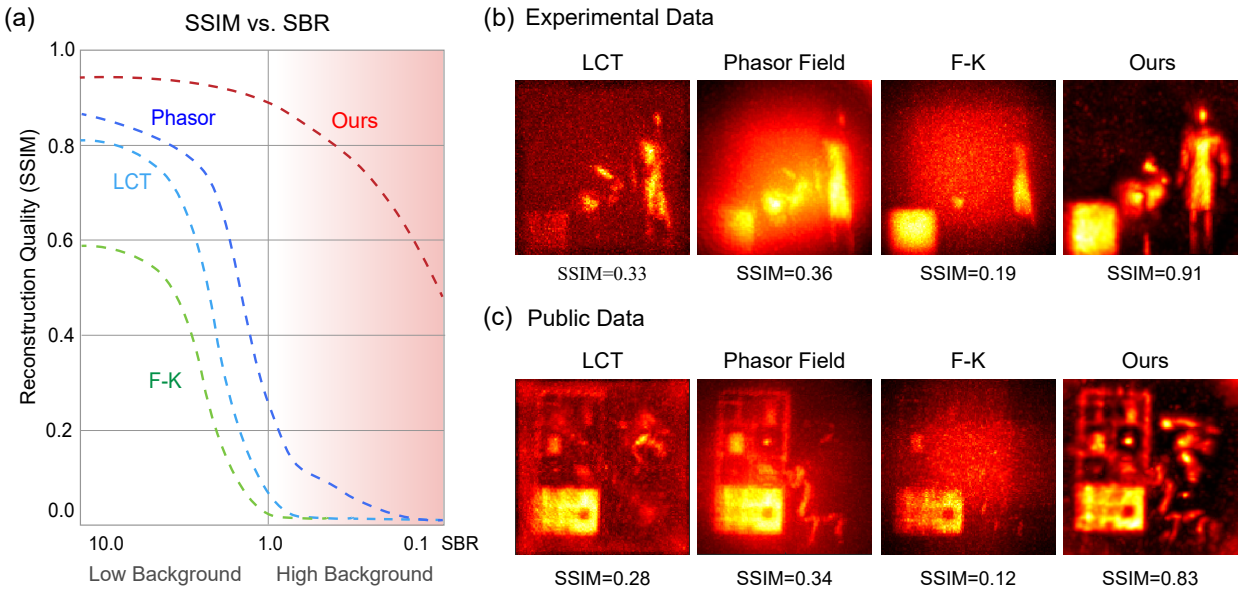


Fig. 4. **Noise robustness.** (a) Quantitative evaluation of reconstruction quality (SSIM) versus signal-to-background ratio (SBR) on simulated data. While baseline methods suffer from a sharp performance collapse as background noise increases, our method demonstrates superior robustness, maintaining high fidelity even in noise-dominated regimes (SBR < 1.0). (b) Experimental reconstructions under strong ambient light (SBR ≈ 1.9). (c) Comparative results on public data [12] with synthetic noise injection (SBR ≈ 1.6). In both scenarios, our method recovers clear geometric features where voxel-based baselines suffer from severe noise artifacts.

using simpler geometries like the Stanford Bunny, showing that as few as 1000 Gaussian primitives suffice for accurate reconstruction.

Based on the extreme compactness, we further evaluate sparse reconstruction performance under both confocal (Fig. 3(c), 3(d)) and non-confocal configurations (Fig. 3(e)). For these experiments, we utilize our own experimental data alongside the public "fk-teaser" confocal dataset [12] (180-minute integration) and the "shelf-target-lighton" non-confocal dataset [11]. We compare our method against LCT [10], Phasor Field [11], and F-K [12] for confocal scenarios, while utilizing FBP [9], Phasor Field [11], and Fast RSD [16] as baselines for the non-confocal setting. As shown in Fig. 3, these voxel-based algorithms suffer from severe aliasing artifacts or fail when the sampling density decreases. In contrast, our method achieves high-fidelity reconstruction of complex NLOS scenes using as few as 24×24 or even 16×16 sampling points. Particularly in the non-confocal setting, we achieve accurate reconstruction using merely $\sim 2\%$ of the full scanning grid, which was previously considered unattainable. This ability to recover reliable scene structure from extremely sparse measurements directly translates into much faster data acquisition, substantially lowering the barrier to fast and large-scale NLOS imaging. (See Supplementary Section 3 for more details.)

B. Noise Robustness

Compared to discrete voxels, our Gaussian field possesses inherent spatial smoothness. When combined with the regularization enforced by our optimization pipeline, transient Gaussian field exhibits significantly enhanced robustness to noise. To validate this, we first perform a quantitative evaluation of reconstruction quality (SSIM) versus signal-to-background ratio (SBR) on simulated data in Fig. 4(a). Representative reconstructions corresponding to different points along the curves are shown in Supplementary Fig.S7. As the background noise increases, baseline methods experience a rapid degradation in performance, ultimately becoming ineffective in low-SBR regimes. In contrast, our method maintains high reconstruction fidelity even under noise-dominated conditions (SBR < 1.0).

We further validate this robustness through experimental imaging under strong ambient illumination. The experimental setup is identical to that shown in Fig. 3(c), except that a 128×128 sampling grid is used and all lights are turned on during acquisition, resulting in a challenging SBR of approximately 1.9. As shown in Fig. 4(b), voxel-based solvers, including Phasor Field, LCT, and F-K, recover only coarse object outlines that are heavily contaminated by noise. By contrast, our approach achieves high-fidelity reconstructions, faithfully capturing the fine geometrical details of both the mannequin and the bunny with sharp boundaries while effectively suppressing noise-induced artifacts. This noise robustness is also shown quantitatively, with our method achieving an SSIM of 0.91, significantly outperforming all baselines. Consistent improvements are further observed on the public "fk-teaser" dataset [12] under synthetic noise injection (Fig. 4(c)). Across all evaluated scenarios, our method consistently delivers superior visual quality and quantitative performance. These results confirm that transient Gaussian field maintains stable and accurate reconstruction even under severe ambient interference. (See Supplementary Section 4 for details).

C. Overcoming Finite-Aperture Constraints

A fundamental challenge in optical NLOS imaging arises from the finite size of the relay wall, which acts as a limited

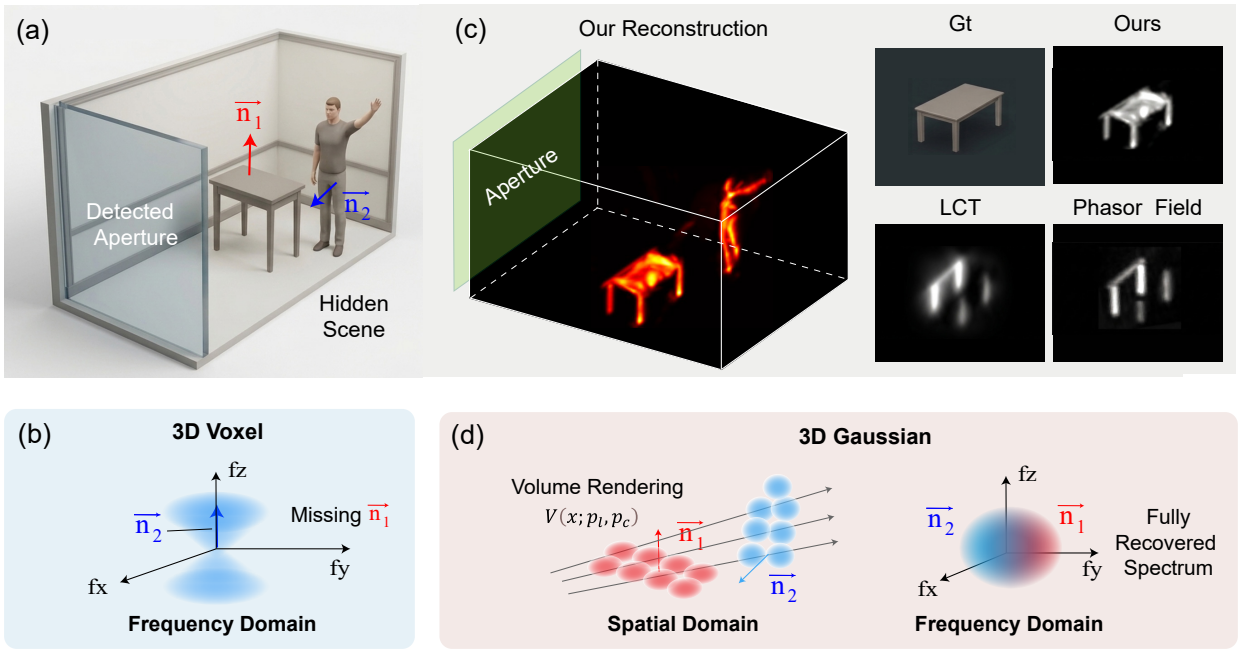


Fig. 5. **Overcoming finite aperture constraints via transient Gaussian field.** (a) The relay wall acts as a finite aperture, restricting angular coverage. (b) This creates a "missing cone" in the frequency domain, causing voxel-based methods to fail on surfaces not facing the wall [21]. (c-d) Our method leverages volume rendering to mix spatial frequencies, thereby allowing full structural information to pass through the finite aperture. This yields a much broader recoverable spectral bandwidth compared to LCT [10] and Phasor Field [11], ensuring the complete recovery of 3D shapes such as the horizontal tabletop.

synthetic aperture. Methods to overcome this limitation have depended on resolution enhancements from clean vertical edges in known positions, which are not generally available [47], [48]. In standard NLOS settings, voxel-based methods commonly adopt an assumption of isotropic visibility, whereby each voxel is presumed visible from all detection points on the wall. Under spatially limited sampling, this simplified forward model restricts the recoverable Fourier-domain support to a confined cone [21], as illustrated in Fig. 5(b). Consequently, surface normals oriented outside the detection aperture produce spatial-frequency components that fall within the spectral "missing cone" and are therefore unrecoverable. This limitation is clearly manifested in the Phasor Field and LCT reconstruction shown in Fig. 5(c). Although the captured transient measurements contain valid photon returns from both the tabletop and the mannequin, only the mannequin facing the relay wall is reconstructed. In contrast, the upward-facing tabletop, whose surface normal is directed outside the aperture, is entirely absent. Its echoes are misinterpreted as noise and suppressed by the reconstruction process. Therefore, in most everyday scenes, voxel-based methods typically yield fragmented, discontinuous reconstructions, offering little help for the perception and understanding of hidden spaces.

Here, we demonstrate that transient Gaussian field effectively circumvents this challenge through a physically rigorous volume rendering model. Unlike voxel-based methods, our framework explicitly computes the view-dependent visibility, $V(\mathbf{x}; \mathbf{p}_l, \mathbf{p}_c)$, for each Gaussian component within the field. In the frequency domain, the multiplication of the reflectance field $\rho(\boldsymbol{x})$ and the visibility term corresponds to a convolution of their spectra, $\mathcal{F}[\rho(\boldsymbol{x})V(\boldsymbol{x}; \boldsymbol{p_l}, \boldsymbol{p_c})] = \mathcal{F}[\rho(\boldsymbol{x})] * \mathcal{F}[V(\boldsymbol{x}; \boldsymbol{p_l}, \boldsymbol{p_c})]$. Because the visibility function exhibits broad spectral support, this convolution effectively blends spatial frequencies from the 'missing cone' into the observable passband, allowing high-frequency geometric components to be reconstructed despite the finite aperture. As demonstrated in Fig. 5(c), our method enables the recovery of the full frequency content of the scene, including surfaces previously invisible to the relay wall, such as horizontal tabletops. In Supplementary Fig.S14, we further validate on public data [21] to confirm this advantage. While baseline methods recover only the two patches facing the detection area, our method successfully reconstructs all three patches. Consequently, even under identical finite-aperture measurements, our field-based framework reconstructs a substantially larger scene coverage than existing approaches. (See Supplementary Section 5 for a detailed theoretical derivation and spectral analysis.)

D. 3D Panoramic NLOS Imaging

Leveraging all these capabilities, we present the first demonstration of 3D panoramic NLOS imaging. By utilizing different relay walls, such as two parallel relay walls (Fig. 6) or the side wall and ceiling of a room (Fig. 7), we can recover the full 3D geometry of the hidden scene. A critical innovation in our framework is the unified volumetric optimization, where we treat disparate transient measurements as constraints on a single, unified set of Gaussian field elements. This differentiable rendering pipeline naturally fuses photon arrival times from multiple views into a consistent 3D geometry.

To validate the feasibility of 3D panoramic imaging, we have conducted two experiments in Fig. 6(b)(c). The two walls are separated by 2 m, each providing a scanning area of 1.95×1.95 m. We evaluate two scenarios: a David statue (0.8

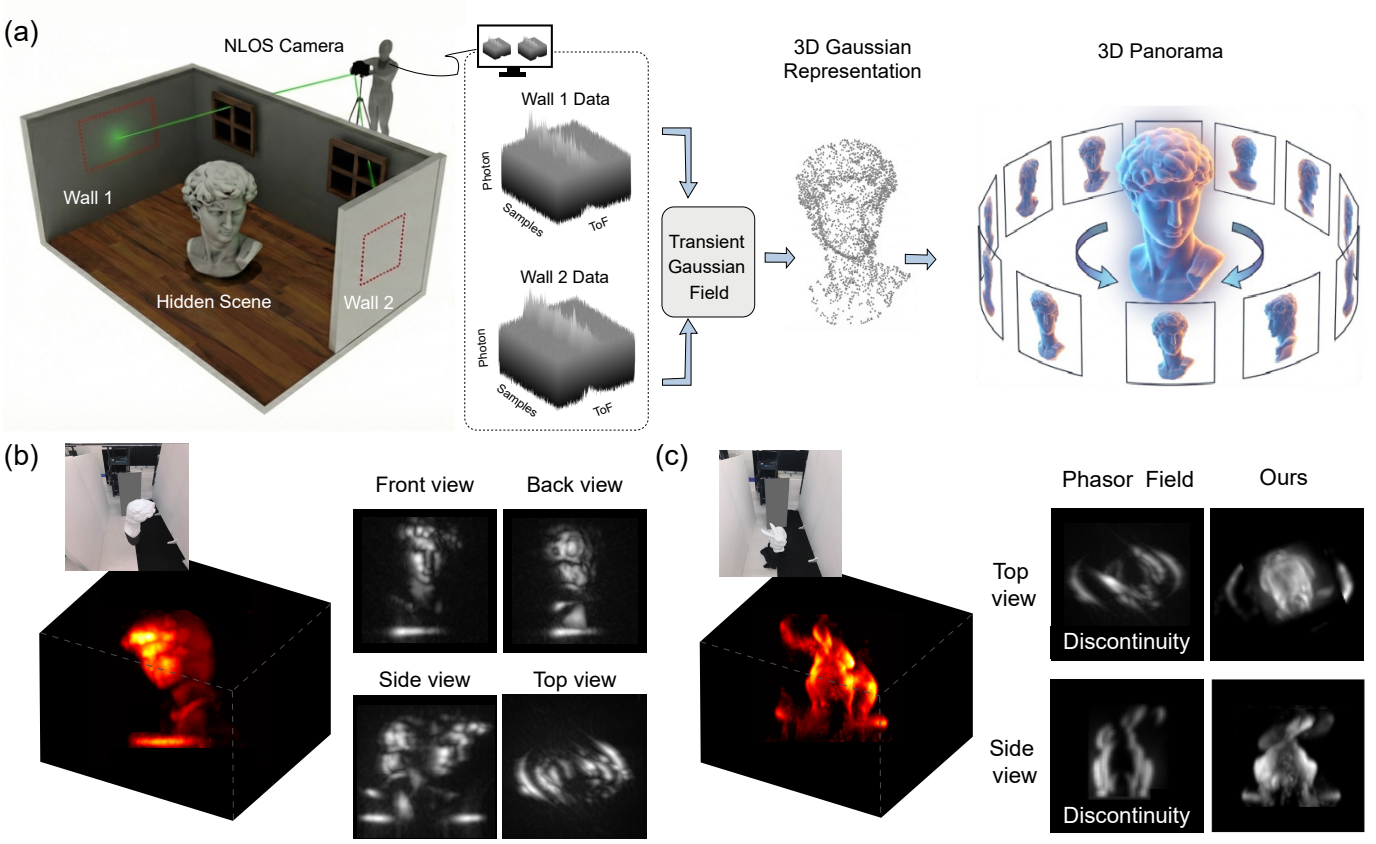


Fig. 6. **3D panoramic NLOS imaging experiments through finite apertures.** (a) Schematic of the proposed panoramic reconstruction framework. The system captures transient signals from limited areas on two parallel walls. These measurements are fused to optimize a unified Gaussian field representation, which enables rendering novel views from arbitrary angles. (b-c) Experimental reconstruction results for the 'David' and 'Bunny' scenes, respectively. Notably, our method successfully achieves high-quality 360° panoramic reconstruction of the hidden objects, whereas voxel-based methods exhibit pronounced discontinuities.

m height) and a Bunny statue (0.6 m height), both positioned centrally facing the detection walls. Unifying the time-of-flight histograms from both walls into a single dataset, we perform volume rendering with 128×128 rays per view, sampling 16 detection points per mini-batch. The model typically converges within 500 iterations. The final results demonstrate that our optimized continuous Gaussian field representation allows for free-viewpoint rendering, generating high-quality, complete 360-degree reconstructions. We also compare our method with the Phasor Field approach under the same experimental settings (see Supplementary Fig.S15). The results demonstrate that our method successfully reconstructs a complete 3D panorama. Importantly, not only are the front and back views accurately recovered, but the previously unobserved top and side views also appear smooth and continuous. In contrast, the Phasor Field reconstruction exhibits noticeable discontinuities in these transition regions. This comparison demonstrates that our method fundamentally extends the recoverable viewing range of NLOS imaging and enables comprehensive 3D perception in complex scenes.

In Fig. 7, we further demonstrate panoramic imaging around a corner for complex everyday scenes, enabling navigation within the hidden space. In a $2 \times 2 \times 2$ m room, there are a desk, a bed, and a person. By probing one side wall and a small visible portion of the ceiling, we can recover the complete hidden space, as shown in Fig. 7(a). More importantly, our reconstructed hidden space is no longer a standard RGB-D image, but an explorable 3D world that can be investigated in 3D space by moving the position of a virtual camera. Fig. 7(c) and (d) show scene images rendered from two different virtual cameras, corresponding to the blue and green camera trajectories in Fig. 7(a), respectively. Both navigate deep into the scene, allowing for the observation of scene details, which is of great significance in applications such as disaster rescue.

VII. Discussion

In this work, we show that the continuous Gaussian field offers a far more effective spatial representation than discrete 3D voxels, capable of dramatically extending the applicability of NLOS systems. As shown in Fig.8, our method not only performs better in standard confocal and non-confocal configurations, but also remains effective in challenging yet practical scenarios, including sparse sampling and severe noise. Furthermore, we overcome the finite-aperture constraints and present the first experimental demonstration of 3D panoramic NLOS imaging. Together, these advances establish a new benchmark for computational periscopy. (A detailed comparison with existing methods is provided in Supplementary Section 7.)

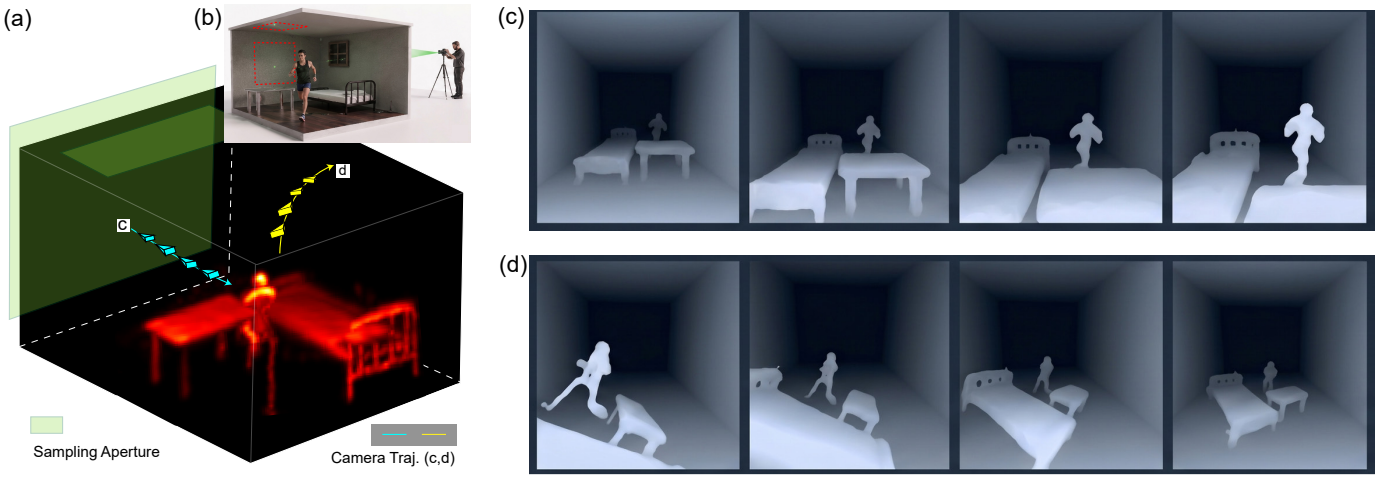


Fig. 7. **Camera roaming and free-viewpoint synthesis within the reconstructed hidden space.** (a-b) Panoramic reconstruction of a complex everyday indoor scene. Our continuous Gaussian field representation supports seamless free-viewpoint navigation, allowing the observer to roam inside the hidden volume and examine scene details from arbitrary angles. (c-d) Synthesized views extracted from two distinct camera trajectories, corresponding to the cyan path (c) and the yellow path (d) illustrated in **a**.

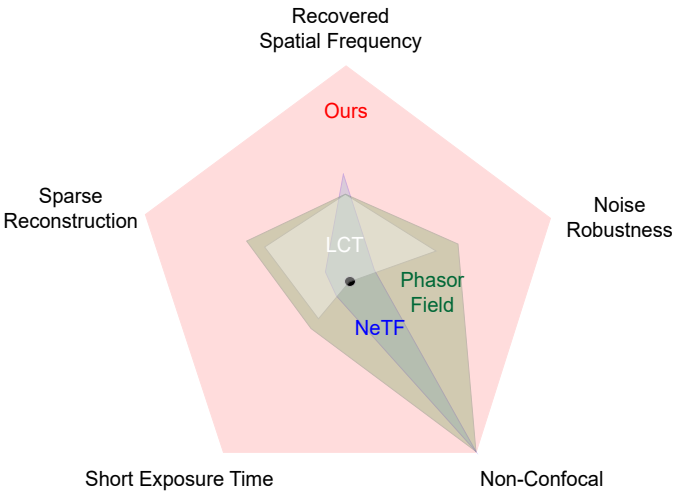


Fig. 8. **Comparison of NLOS imaging methods.** Our method demonstrates comprehensive superiority in reconstruction capability compared to other approaches.

Although the Gaussian field reduces the representation parameters, the reliance on traversing N^2 scanning points dictates a computational complexity of $O(N^4)$ for both confocal and non-confocal settings. While this is computationally heavier than the $O(N^3 \log N)$ complexity of LCT [10] in the confocal scenario, it remains more efficient than the $O(N^5)$ complexity of back-projection [9] in the non-confocal scenario. Future work can integrate deep learning [35], [36], [49] to accelerate inference while preserving the advantages of the continuous field representation. It will be interesting to extend our framework to dynamic scenes by leveraging 4D space-time representations [39], [50] that natively capture scene dynamics. Beyond NLOS imaging, our framework offers a generalizable solution to inverse problems involving wave propagation. We believe these principles hold promise for other related fields, such as acoustic imaging [51], [52], seismic exploration [53], and medical tomography [54].

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